

RESUME

RONY HALDER

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Github: updating ..



Senior Hardware / PCB Engineer with 10+ years of experience developing electronic products from concept through prototype, validation, manufacturing and deployment. Strong hands-on background in high-density multi-layer PCB design, schematic capture, mixed-signal and RF hardware, power electronics, embedded hardware and low-level firmware integration. Experienced with Altium Designer and Mentor Graphics, component selection, BOM optimization, DFM, supplier coordination, hardware troubleshooting, bench testing, root-cause analysis and technical documentation. Relevant power-electronics experience includes SMPS, DC-DC converters, inverters, battery charging/protection, Li-ion BMS, wireless power and power-management systems.

CAREER OBJECTIVE:

Seeking a Senior Hardware Engineer or Embedded Systems Engineer role where I can apply a decade of experience in multi-layer PCB design, embedded development, and full-lifecycle product engineering to create reliable and innovative tech. I am looking to contribute to challenging R&D projects—such as wireless systems, battery management, and industrial automation—where I can solve complex technical problems and support products all the way from initial schematic design through to manufacturing and deployment. Ultimately, I want to use my engineering skills to build meaningful technology that helps make this world a better place.

SKILLS:

Hardware & PCB Design

- **CAD Tools:** Altium Designer, Mentor Graphics, Eagle
- **PCB Expertise:** High-speed & multi-layer PCB design, mixed-signal RF design, schematic capture, component selection, design modification(multilayer stack impedance matching)
- **Manufacturing:** DFM; PCB fabrication processes; quality inspection; hardware validation; compliance testing; supplier coordination; production support
- **Power Electronics:** SMPS design, ferrite core transformer design, Battery Management Systems (BMS), wireless power
- **Hardware:** ARM Cortex/STM32; AVR; RISC-V/WCH; ESP32; embedded C/C++; MicroPython; UART; SPI; I2C; One-Wire; Bluetooth; RF
- **Hardware Troubleshooting:** System-level and component-level troubleshooting; prototype testing; bench testing; fault isolation; root-cause analysis; design modifications and validation

Power Electronics:

- **SMPS:** Switchmode power supply design, prototyping , implementation and testing
- **DC-DC converters:** High speed Buck/ Boost/SEPIC converters, MPPT charge controllers design, implementation and testing
- **Inverters:** IPS/UPS design, implementation and testing. (Working on solar microgrid)

Embedded Systems & Firmware

- **Architectures:** ARM Cortex (STM32), AVR, RISC-V (CH32V series), ESP32, Arduino etc.
- **Programming Languages:** C, C++, MicroPython

- **IDEs & Toolchains:** STM32CubeIDE, Keil uVision, TrueStudio, Atmel Studio, MounRiver Studio, winAVR, Thonny, Visual Studio Code etc.
- **Hardware Protocols:** UART, SPI, I2C, One-Wire, RF, Bluetooth, IoT system architecture

Industrial Automation & Control

- **Systems:** PLC, HMI, VFD, Power Factor Improvement (PFI) plants, control systems, manual automation
- **Field Engineering:** On-site factory automation troubleshooting, debugging, and system integration

IT, Networking & Systems (Supporting Expertise)

- **Infrastructure:** LAN/WAN networking, computer hardware assembly, system troubleshooting, multi-OS environments (Windows, Linux, macOS)
- **Software & Web:** WordPress, HTML5, CSS3, JavaScript

EMPLOYMENT HISTORY:

Company	:	THT-Space Electrical Ltd.	(Current, from Jan, 2024)
Location	:	Uttara EPZ, Nilpahamari	
Designation	:	Embedded Software Engineer	
Responsibilities	:	• Designing complex multi-layer mixed-signal and RF circuits using Altium Designer and Mentor Graphics. • Authoring comprehensive technical documentation, hardware specifications, and production reports. • Drawing, editing, and optimizing detailed schematics and high-density PCB layouts. • Leading the Hardware team and collaborate between hardware and software team. • Conducting component selection, sourcing, and Bill of Materials (BOM) optimization for R&D projects. • Prototyping, troubleshooting, testing, and debugging hardware systems to isolate and resolve technical flaws. • Managing technical communication, component sourcing, and Design for Manufacturing (DFM) alignment with Chinese suppliers. • Overseeing strict quality inspection protocols for components, prototypes, and final production assemblies.	
Achievements	:	• Wireless Thermal Printers(Hardware) • Dot Printers(Hardware) • Wireless Power Bank(Hardware) • Face Attendance Machine(Hardware) • Robotic Arm(Hardware) • BMS for Li-ion Batteries(Hardware)	
Company	:	Exact Engineering Solutions Ltd.	(Mar, 2023-May 2023)
Location	:	TTI, BITAC, Tejgaon, C/A, Dhaka.	
Designation	:	Senior Embedded System Engineer	

- Responsibilities :
- Designing complex multi-layer mixed-signal circuits with a strong focus on signal and power integrity.
 - Authoring detailed technical documentation, engineering reports, and production-ready hardware specifications.
 - Drawing, editing, and optimizing high-density schematics and PCB layouts.
 - Developing and debugging efficient, low-level firmware using C/C++ for microcontroller-based systems.
 - Managing component selection, verifying component footprints, and optimizing the Bill of Materials (BOM).
 - Prototyping, troubleshooting, testing, and debugging hardware platforms to isolate and resolve system-level flaws.
 - Reverse engineering existing electronic hardware to analyze circuitry, reconstruct schematics, and extract firmware logic.
 - Updating old designs to modern components, replacing obsolete parts, and redesigning boards to improve efficiency and lower manufacturing costs.
- Achievements :
- Network Jammer(Hardware)
 - Plastic Scanner(Hardware + Firmware)
 - Rain Chamber(Hardware + Firmware)

Company	:	Minister Hi-tech Park Electronics Ltd.	(Apr 2022-Nov, 2022)
Location	:	Narayanpur, Trishal, Mymensingh.	
Designation	:	Electrical R&D Engineer	
Responsibilities	:	• Designing, simulating, and analyzing complex electronic products and multi-layer circuits for mass production. • Authoring detailed technical documentation, design verification reports, and manufacturing specifications. • Drawing, editing, and optimizing high-density schematics and PCB layouts using industry-standard CAD tools. • Developing and debugging robust, low-level firmware in C/C++ for a wide range of microcontroller architectures. • Conducting component selection, verifying mechanical footprints, and optimizing the Bill of Materials (BOM) for reliability and cost-efficiency. • Overseeing strict quality inspection, hardware validation, and compliance testing for electronic equipment and assemblies. • Prototyping, troubleshooting, bench testing, and debugging hardware and firmware to isolate and resolve system-level flaws. • Managing technical communication, component sourcing, and Design for Manufacturing (DFM) alignment with international suppliers.	

- Achievements :
- Digital Thermostate(Hardware + Firmware)

Company	:	Furev Technology	(Jan,2015-Jun, 2020)
Location	:	Joar Shahara, Kuril, Dhaka	
Designation	:	Embedded System Engineer	

- Responsibilities :
- Designing, analyzing, and optimizing both analog and digital circuits for stable and efficient performance.
 - Authoring comprehensive technical documentation, design verification reports, and manufacturing guidelines.
 - Drawing, editing, and routing high-density schematics and multi-layer PCB layouts using CAD tools.
 - Developing and debugging robust, low-level firmware in C/C++ to control microcontroller-based systems.
 - Conducting rigorous component selection, verifying mechanical footprints, and optimizing the Bill of Materials (BOM).
 - Prototyping new hardware, troubleshooting system-level flaws, and performing bench testing and debugging.
 - Reverse engineering existing electronic hardware to analyze architectural layouts, reconstruct schematics, and update legacy systems.
 - Overseeing strict quality inspection protocols, component validation, and hardware compliance testing for final assemblies.
- Achievements :
- Audio Circuits (Active Crossovers + Amplifiers)
 - Dental Chair(Hardware + Firmware)
 - Industrial Capping Machine(Hardware Manual Automation)
 - Digital Timer for Refrigerator(Hardware + Firmware)
 - Lead Acid Battery Charging and Protection(Hardware)
 - SMPS(Hardware)

AREAS OF EXPERTISE:

- **Hardware Architecture:** Analog and digital circuit design, simulation, and analysis.
- **High-Speed PCB Design:** Advanced schematic capture, multi-layer routing, and high-density layouts using Altium Designer, Mentor PADS, Eagle, KiCAD, and EasyEDA.
- **Firmware Development:** Writing robust low-level code in C/C++ and MicroPython across diverse architectures including ARM Cortex (STM32), AVR, PIC, RISC-V (WCH), ESP32, and Arduino.
- **Power Electronics:** Designing and optimizing power management systems, including inverters, converters, and industrial power supplies.
- **Prototyping & Diagnostics:** Hands-on hardware prototyping, bench testing, component-level debugging, and advanced system troubleshooting.
- **PCB Fabrication & DFM:** Practical experience with screen printing, photolithography, and CNC milling processes to bridge design with physical manufacturing.
- **Industrial Automation:** Designing and maintaining manual and PLC-based automated control systems for factory environments.
- **Technical Documentation:** Compiling detailed hardware specifications, design verification reports, and manufacturing documentation.

Recent Projects:

- **Solar Micro Grid:** (MPPT controller, battery management system, inverters, active PFC module)
- **Liquid vending machine:** ESP32 HMI, Motor driver, load sensor, one wire communication, Bluetooth
- **Thermal Printers:** RISC-V MCU, print head, motor driver, BMS, WiFi, BLE
- **AI Camera:** RISC V AI chip, DVP camera, MIPI Camera, PD protocol

- **Robotic Arms** :RISC-V MCU, motor driver, BLE

PROFESSIONAL COURSES:

Course : Advanced Industrial Automation (PLC, HMI, VFD, Servo Drive)

Institute : Bangladesh Automation.

Duration : 1 year

LANGUAGE PROFICIENCY:

English, Bengali (Fluent in writing, reading and speaking).

EDUCATIONAL QUALIFICATION:

Level	Concentration	Institute	CGPA
Bachelor of Science	BSc in EEE	Sonargaon University	3.93 Out of 4
Higher Secondary School Certificate	Science	Notre Dame College	4.30 Out of 5
Secondary School Certificate	Science	Gawair Adarsha High School	5 Out of 5

PERSONAL INFORMATION:

Father : Amal Halder

Mother : Usha halder

Gender : Male.

Religion : Christianity

Marital Status : Single.

Date of Birth : 15th January, 1990.

Nationality : Bangladeshi (By Birth).

PERMANENT ADDRESS:

Village: Lakhirpar, Police Station: Kotalipara, Post Office: Radhaganj, District: Gopalganj.

REFERENCE:

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DECLARATION:

I hereby declare that all the information given above is true to the best of my knowledge.

A handwritten signature in dark ink, appearing to be 'Rony Halder', written over a horizontal line.

Rony Halder